

ACS 597: Noise Control Applications

Vibration isolation

1 Washing machine mount design

You are designing a mount for an industrial washing machine to be installed on a ship. The machine has mass 100 kg. The dynamic force on the machine can be modeled as an added imbalanced mass of 1 kg at a radius of 0.5 m. The washing machine's operating speed is 300 RPM.

The maximum dynamic force that can be applied to the foundation is 100 N. When the washing machine is loaded (10 kg added), the static displacement must be less than 1 cm.

- (a) Write an expression for the amplitude of the dynamic force that the imbalanced mass applies to the washing machine, as a function of RPM. Plot the amplitude of the force for 0–350 RPM. What is the force amplitude and frequency at 300 RPM?
- (b) What is the minimum mount stiffness required to meet the static displacement criterion? What is the natural frequency of a mount with this stiffness, if the total mass of the system is the washing machine's mass plus the added mass?
- (c) Using this stiffness, plot the transmission loss of the mount as a function of frequency for several damping ratios ranging from 0.001 to 1.
- (d) Multiply the transmission loss by the force amplitude to get the force applied to the foundation as a function of RPM. Is there a damping ratio for which the force is less than the requirement across the whole operating range?
- (e) Choose the best damping ratio from the previous part. Calculate the viscous damping coefficient, C , and plot the admittance magnitude of the mount as a function of frequency.
- (f) Multiply the admittance by the force to get the displacement amplitude of the washing machine as a function of frequency. Plot this displacement as a function of RPM. What is the maximum displacement of the washing machine? Do you think this is a reasonable amount of displacement for a washing machine under operating conditions?

2 Washing machine two-stage mount design

You have decided that the displacement you calculated at the end of the previous problem is too high. Design a two-stage mount for the washing machine. The additional mass of the mount base (sometimes called the “raft”) is limited to 100 kg.

- (a) Draw a diagram of the two stage mount, labeling all parameters and variables. You can incorporate damping either with viscous damping elements or by making the spring stiffnesses complex.
- (b) Choose initial values for the new stiffness and damping coefficients.
- (c) Calculate the admittance of both the washing machine and the raft for a force on the washing machine. You may use the provided MATLAB function `nDOF_direct_solution` for this calculation.
- (d) Calculate the displacement of both masses using the applied force amplitude from Problem 1 and the admittance. Make sure you only apply the force to the washing machine and not to the raft.
- (e) Calculate the force that the raft applies to the ground as a function of frequency.
- (f) Adjust the various mount parameters and attempt to tune the system such that the maximum dynamic displacement of the washing machine is as low as possible while keeping the force transmitted to the floor below the requirement.
- (g) Compare the displacement and transmitted force of your final design to the design with your initial values. What did you change to optimize the mount design?

3 Washing machine dynamic vibration absorber design

Instead of a two-stage mount, you decide to add a dynamic vibration absorber (DVA) to the washing machine, with the goal being to reduce the transmitted force as much as possible at the operating condition. The additional DVA mass is limited to 15 kg. Take a similar approach to the two stage mount, but with the following changes:

- The washing machine is now connected to the floor, with the DVA attached to the washing machine. Notice that the force is still on the washing machine.
- The principle goal is to reduce the displacement and transmitted force at the operating condition. For this problem, we will assume that a ramp-up in RPM happens quickly enough to not be a concern. However, all plots should still be functions of frequency.

4 Washing machine abuse testing

The washing machine is required to withstand a certain amount of abuse. While at its operating condition, a 5 kg brick is suddenly introduced to the washing machine.

- (a) Model this system assuming the load was perfectly balanced before the introduction of the brick, so the forcing function suddenly goes from zero to the imbalanced load forcing.
- (b) Using your design from Problem 1, plot the predicted displacement time history of the washing machine.
- (c) Approximately how long does it take to reach a steady-state response?
- (d) What is the maximum transient displacement and acceleration of the washing machine?